PaperMiner Sample Paper

Abstract
This is a sample paper used to test PaperMiner. We study whether reading assistants improve comprehension of academic papers. We collected 120 participants and measured comprehension, time on task, and satisfaction. Results show that AI-assisted reading improves comprehension by 18 percent while reducing reading time. We discuss limitations including the short duration of the study and the use of a single discipline.

1 Introduction
Reading academic papers is difficult. Prior work has explored summarization and question answering. In this paper we build a reading assistant and evaluate it in a controlled experiment.

2 Method
We recruited 120 graduate students. Participants were randomly assigned to a control group and an AI-assisted group. The AI assistant provided summaries and answered questions about the paper. Comprehension was measured with a 20-item quiz.

3 Results
The AI-assisted group scored 18 percent higher on the comprehension quiz and spent 23 percent less time reading. Satisfaction was also higher. No significant difference was found for recall after one week.

4 Discussion
The results suggest that AI reading assistants can improve short-term comprehension. However, long-term retention did not improve, which may be due to cognitive offloading. Limitations include a small sample and a single discipline. Future work should study longer interventions.

5 Conclusion
AI-assisted reading improves short-term comprehension and efficiency. Long-term effects remain uncertain.